

Beyond the Matrix Sign: Quadratic Spectral Descent

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Abstract

Muon can be interpreted as optimizing a linear local objective over a spectral-norm ball. This gives a matrix-sign update that preserves the singular directions of the gradient and assigns the same magnitude to all active singular modes. We ask whether these two properties remain optimal when local curvature is taken into account. To answer this question, we keep Muon’s spectral-norm constraint unchanged and replace the linear local model with a quadratic one. We call the resulting method *Quadratic Spectral Descent* (QSD). We show that curvature can change both the singular values and the singular directions of the optimal update. To make QSD practical, we approximate curvature with Kronecker-factored statistics and solve the constrained quadratic with a small number of Frank–Wolfe steps, each of which has a closed-form matrix-sign subproblem. We further provide an optimality certificate, a comparison with Muon under the same quadratic surrogate, and an $O(1/K)$ convergence rate for the inner solver. Experiments on GPT pre-training show that QSD consistently improves validation loss over Muon and recent Muon variants, and reduces wall-clock training time by up to 8.49% at matched validation loss.

1 Introduction

As language models and pre-training datasets continue to scale, the computational cost of pre-training has grown substantially, making training efficiency increasingly important [14]. While AdamW [18] remains widely used for large-scale pre-training, recent work has explored matrix-aware optimization as an alternative to coordinate-wise adaptive methods. A representative example is Muon [13, 17], which applies matrix orthogonalization to parameter updates and has shown strong performance and favorable scalability in language-model training [17, 25, 30].

Muon also has a simple constrained-optimization interpretation. Let $G = U\Sigma V^\top$ denote the gradient matrix, or its momentum-smoothed counterpart in practice. Ignoring practical rescaling and weight decay, the Muon direction is

$$D_M = -\rho \operatorname{msgn}(G) = -\rho UV^\top, \quad (1)$$

which exactly solves

$$\min_{\|D\|_2 \leq \rho} \langle G, D \rangle. \quad (2)$$

Thus, Muon gives the optimal update for a *linearized* local objective over a spectral-norm ball [23, 2]. The solution has two simple properties: it preserves the singular directions of the gradient and assigns the same magnitude ρ to every active singular mode.

For a finite parameter update, however, two directions with similar first-order decrease can incur substantially different second-order costs. It is therefore natural to ask whether the two properties above are still optimal when local curvature is taken into account.

Recent work has explored curvature, preconditioning, and non-flat spectral structure in Muon-style optimization [26, 6, 5, 32, 33, 35, 3]. Some methods use curvature through preconditioning or modified optimization geometries, while others directly change the singular spectrum of the update. [26] instead studies the optimal update under an isotropic curvature model. Under this model, curvature can lead to non-uniform singular values, but the singular directions remain aligned with those of the gradient. It is therefore still unclear how anisotropic curvature changes the optimal update under Muon’s original spectral-norm constraint.

In this paper, we study this question by keeping the feasible set unchanged and replacing the linear local model with a quadratic one:

$$\min_{\|D\|_2 \leq \rho} \left\{ \langle G, D \rangle + \frac{\eta}{2} \text{vec}(D)^\top \mathcal{C} \text{vec}(D) \right\}, \quad (3)$$

where $\mathcal{C}$ is a positive-semidefinite curvature operator and η is the learning rate. When $\mathcal{C} = 0$, this reduces exactly to the Muon problem. When curvature is nonzero, the update must balance first-order decrease with second-order cost. We call this formulation **Quadratic Spectral Descent (QSD)**.

The main difficulty is that Eq. (3) does not in general have the closed-form matrix-sign solution of Muon. Moreover, the curvature must be estimated efficiently during training. We use K-FAC [19, 8] to approximate the curvature and a small number of Frank–Wolfe steps [7, 11] to solve the constrained quadratic. Each Frank–Wolfe linear subproblem over the spectral-norm ball still has a closed-form matrix-sign solution. With zero initialization, the first Frank–Wolfe atom is exactly the Muon direction, while later atoms depend on curvature-corrected gradients.

The quadratic model changes both properties of the Muon update. First, even when the gradient and curvature share the same singular modes, the optimal singular values are generally not equal. Directions with larger curvature receive smaller updates unless the gradient is sufficiently strong. Second, under more general curvature, the optimal singular directions need not coincide with those of the gradient. Therefore, neither the flat singular spectrum nor the gradient-aligned singular directions are required by the spectral-norm constraint. They are optimal for the linear problem solved by Muon, but are not generally optimal for the quadratic problem.

Our contributions are summarized as follows:

- **Quadratic Spectral Descent.** We extend Muon’s linear local problem to a quadratic one while keeping the same spectral-norm constraint. We show that curvature can make the optimal singular values non-uniform and can also change the optimal singular directions.
- **Efficient optimization of QSD.** We develop an efficient solver for QSD using Kronecker-factored curvature and a small number of Frank–Wolfe steps. The first Frank–Wolfe step recovers the Muon direction, while later steps refine it using curvature information.
- **Optimization guarantees and empirical validation.** We provide a computable optimality certificate, a comparison with Muon under the same quadratic surrogate, and an $O(1/K)$ convergence rate for the inner solver. Experiments on GPT pre-training show consistent improvements over Muon and recent Muon variants, with up to 8.49% lower wall-clock training time at matched validation loss.

1.1 Related Work

Muon and spectral-norm optimization. Muon can be interpreted as normalized steepest descent under the spectral norm, or equivalently as linear optimization over a spectral-norm ball [2, 4, 24, 23]. This viewpoint explains the matrix-sign update as the exact solution of a linear minimization oracle over the spectral-norm ball and connects Muon to more general Linear Minimization Oracle (LMO)-based matrix optimization methods such as Scion [23]. Related analyses have studied the geometry, convergence, and implicit properties of Muon-style updates. Our work starts from the same spectral-norm geometry, but asks what update is preferred when the local objective is extended from linear to quadratic.

Curvature-aware and spectrally adaptive Muon variants. Recent work has incorporated curvature, preconditioning, or adaptive spectral structure into Muon-style optimization [27]. Newton-Muon [6] derives a curvature-informed matrix-sign update from a local quadratic surrogate. FISMO [33] introduces second-order information through a Fisher-dependent trust-region geometry, while Mousse [35] uses curvature-aware whitening to adapt the spectral geometry of the update. GO-MUON [3] adapt the spectral geometry using curvature-dependent transformations. Other approaches [5, 32, 31, 21] modify or reshape the singular spectrum of Muon-style updates directly, thereby relaxing the uniformly flattened spectrum produced by a single matrix sign. Recent analysis also attributes part of Muon’s advantage over Adam to a smaller second-order curvature penalty, suggesting that local curvature can materially affect the quality of matrix-sign updates [29]. These works demonstrate several ways in which curvature or spectral anisotropy can improve matrix-based optimization. Our perspective differs in that we retain Muon’s original spectral-norm ball and directly optimize a quadratic local model over the same set, allowing both the singular amplitudes and directions to adapt beyond a single matrix-sign update.

Structured curvature and matrix preconditioning. A large body of work exploits matrix structure to obtain tractable second-order or adaptive updates. K-FAC [19, 8] approximates layerwise curvature using Kronecker-factored statistics, while Shampoo [9, 20] and SOAP [28] construct matrix-valued adaptive preconditioners from structured second-moment information. These methods typically use the resulting matrix statistics to transform or precondition a gradient update. In our method, Kronecker-factored statistics instead specify the curvature term of a local quadratic objective, which is then optimized directly under a spectral-norm constraint. We use Frank–Wolfe optimization for this constrained problem because its linear minimization oracle over the spectral-norm ball remains available in closed form as a matrix-sign operation.

2 Preliminaries and Problem Formulation

We consider the optimization of a matrix-valued parameter $W \in \mathbb{R}^{m \times n}$ with objective $f(W)$. Let $G = \nabla_W f(W)$ denote the gradient with respect to W . We write a single parameter update as

$$W^+ = W + \eta D, \quad (4)$$

where $D \in \mathbb{R}^{m \times n}$ is the update direction and $\eta > 0$ is the learning rate. Throughout the paper, $\|\cdot\|_2$ denotes the spectral norm, $\|\cdot\|_*$ the nuclear norm, $\|\cdot\|_F$ the Frobenius norm, and $\langle X, Y \rangle = \text{tr}(X^\top Y)$ the Frobenius inner product.

Building on the constrained interpretation of Muon in Eq. (2), we extend the same spectral-norm geometry from a linear to a quadratic local model.

2.1 Quadratic Spectral Descent (QSD)

For a finite update, directions with similar first-order descent can incur substantially different second-order costs. Let $H_W = \nabla_{\text{vec}(W)}^2 f(W)$ denote the Hessian and define $\mathcal{H}[D] = \text{unvec}(H_W \text{vec}(D))$. A second-order Taylor expansion gives

$$f(W + \eta D) = f(W) + \eta \langle G, D \rangle + \frac{\eta^2}{2} \langle D, \mathcal{H}[D] \rangle + \mathcal{R}_3(D). \quad (5)$$

As the Hessian is expensive to apply, we therefore use a positive-semidefinite curvature operator $\mathcal{C}$ instead. After removing the constant term and dividing by η , we obtain

$$q_\eta(D) = \langle G, D \rangle + \frac{\eta}{2} \langle D, \mathcal{C}[D] \rangle. \quad (6)$$

We define Quadratic Spectral Descent (QSD) as

$$\boxed{D^\star \in \arg \min_{\|D\|_2 \leq \rho} q_\eta(D).} \quad (7)$$

When $\mathcal{C} = 0$, QSD reduces exactly to the Muon problem.

3 Practical Quadratic Spectral Descent

The QSD problem is only useful if its curvature term can be estimated and the constrained quadratic can be solved efficiently. We address these two issues with K-FAC and Frank–Wolfe, respectively.

3.1 Kronecker-Factored Curvature

Consider a linear layer $h = Wa$, where $W \in \mathbb{R}^{m \times n}$ and $a \in \mathbb{R}^n$. Let $\delta = \nabla_h \ell \in \mathbb{R}^m$. For a single example, the gradient with respect to W has the outer-product form $\nabla_W \ell = \delta a^\top$. The corresponding empirical-Fisher block is $C_{\text{EF}} = \mathbb{E}[(aa^\top) \otimes (\delta\delta^\top)]$. We use the empirical-Fisher factors as an inexpensive positive-semidefinite structured curvature proxy, without assuming that the empirical Fisher is an exact approximation to the GGN or Hessian [15]. Its overall scale is calibrated against the GGN directional curvature in Section 3.3. K-FAC approximates this block by factorizing the expectation [19, 8]:

$$\widehat{C}_{\text{KFAC}} = A \otimes B, \quad A = \mathbb{E}[aa^\top], \quad B = \mathbb{E}[\delta\delta^\top]. \quad (8)$$

Here A captures input-side second-order statistics and B captures output-gradient statistics. The development below only requires $A \succeq 0$ and $B \succeq 0$, so analogous factorizations can also be used for other positive-semidefinite curvature models.

Using $\text{vec}(BDA) = (A^\top \otimes B) \text{vec}(D)$, and the symmetry of A , the corresponding matrix-valued curvature action is $\widehat{\mathcal{C}}[D] = BDA$.

Importantly, K-FAC is used here to define a local quadratic *objective*, rather than merely to construct a preconditioned gradient. We optimize this curvature-aware objective directly over the spectral-norm ball.

3.2 Frank–Wolfe Solver

The QSD problem does not in general have a closed-form solution. We therefore use Frank–Wolfe [7, 11] to solve it approximately. At each step, Frank–Wolfe linearizes the quadratic objective at the current iterate, solves the resulting linear problem over the same spectral-norm ball, and moves toward its solution. Although the original problem is quadratic, each linear subproblem still has a closed-form matrix-sign solution.

Let D_s denote the current inner iterate. The gradient of the quadratic surrogate with respect to D_s is

$$R_s = \nabla_D \widehat{q}_\eta(D_s) = G + \eta B D_s A. \quad (9)$$

The second term corrects the original gradient G according to the curvature at the current iterate. Frank–Wolfe then solves

$$S_s \in \arg \min_{\|S\|_2 \leq \rho} \langle R_s, S \rangle, \quad S_s = -\rho \text{msgn}(R_s). \quad (10)$$

The new candidate is obtained by moving toward this feasible atom:

$$D_{s+1} = (1 - \gamma_s)D_s + \gamma_s S_s, \quad \gamma_s \in [0, 1]. \quad (11)$$

For the quadratic surrogate, γ_s can be obtained by exact line search. Letting $\Delta_s = S_s - D_s$, we have

$$\gamma_s = \text{clip}_{[0,1]} \left(\frac{\langle R_s, D_s - S_s \rangle}{\eta \text{tr}(\Delta_s^\top B \Delta_s A)} \right), \quad (12)$$

whenever the denominator is positive. With $D_0 = 0$, we have $R_0 = G$, and hence the first Frank–Wolfe atom is exactly the Muon direction. Subsequent atoms use the curvature-corrected gradient R_s and progressively refine this initial update.

3.3 Practical QSD

The derivation above assumes fixed curvature factors. In practice, we estimate the K-FAC factors from minibatch statistics and update them periodically during training.

Factor estimation. At a factor-refresh iteration t , using a sample subset $\mathcal{S}_t$, we estimate for each layer ℓ

$$\hat{A}_{t,\ell} = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} a_{i,\ell} a_{i,\ell}^\top, \quad \hat{B}_{t,\ell} = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \delta_{i,\ell} \delta_{i,\ell}^\top, \quad (13)$$

and maintain exponential moving averages

$$A_{t,\ell} = \beta_1 A_{t-j,\ell} + (1 - \beta_1) \hat{A}_{t,\ell}, \quad B_{t,\ell} = \beta_1 B_{t-j,\ell} + (1 - \beta_1) \hat{B}_{t,\ell}, \quad (14)$$

where j is the factor-refresh interval. Between refreshes, the most recent factors are reused. Details could be found in Appendix C.

Curvature calibration. To mitigate the possible scale mismatch in the Kronecker approximation, we introduce a per-layer calibration coefficient $\alpha_{t,\ell} > 0$. The instantaneous calibration $\hat{\alpha}_{t,\ell}$ is obtained by matching the K-FAC and generalized Gauss–Newton (GGN) directional curvatures along the recent update direction. To reduce noise, the instantaneous ratio is clipped before being incorporated into an exponential moving average:

$$\alpha_{t,\ell} = \beta_\alpha \alpha_{t-k,\ell} + (1 - \beta_\alpha) \text{clip}(\hat{\alpha}_{t,\ell}, \alpha_{\min}, \alpha_{\max}). \quad (15)$$

The calibration rescales only the Kronecker-factored term, damping and the curvature inflation factor τ are applied separately. Details are given in Appendix D.

Even after calibration, the quadratic surrogate can still differ from the practical finite-step training objective. We therefore apply a mild curvature inflation factor $\tau > 1$ to make the update more conservative in directions where curvature already plays a significant role. The effect of this inflation is analyzed in Section 5.3. We use $\tau = 1.5$ throughout our experiments.

Together with isotropic damping, the practical curvature operator is

$$\hat{\mathcal{C}}_{t,\ell}[D] = \tau(\alpha_{t,\ell} B_{t,\ell} D A_{t,\ell} + dD), \quad (16)$$

where $d \geq 0$ is the damping coefficient.

Practical Frank–Wolfe update. For clarity, we suppress the layer index ℓ below. As in Muon, we use the momentum-smoothed matrix M_t in place of the instantaneous gradient. The surrogate gradient at the s -th Frank–Wolfe iterate is therefore

$$R_{t,s} = M_t + \eta_t (\tau(\alpha_t B_t D_{t,s} A_t + dD_{t,s})). \quad (17)$$

Since consecutive inner problems change gradually, we warm-start the solver from the previous solution,

$$D_{t,0} = D_{t-1,K}, \quad (18)$$

with $D_{0,0} = 0$. The warm start remains feasible and allows a small number of Frank–Wolfe steps to track the evolving QSD solution. Algorithm 1 summarizes the resulting practical update.

4 Understanding How Curvature Changes Muon Updates

Muon assigns the same magnitude to all active singular modes and preserves the singular directions of the gradient. These two properties are optimal for the linear problem, but it remains unclear whether they are still optimal once curvature is taken into account. We study the two questions separately. We first ask whether curvature changes the singular amplitudes even when the gradient and curvature share the same modes, and then ask whether curvature can also change the preferred singular directions.

Algorithm 1 Practical QSD with K-FAC and Frank–Wolfe

Require: Momentum M_t , factors A_t, B_t , calibration α_t , inflation τ , damping d , learning rate η_t , spectral radius ρ , FW steps K

```
1: if  $t = 0$  then
2:    $D_{t,0} \leftarrow 0$ 
3: else
4:    $D_{t,0} \leftarrow D_{t-1,K}$ 
5: end if
6: for  $s = 0, \dots, K-1$  do
7:    $R_{t,s} \leftarrow M_t + \eta_t (\tau(\alpha_t B_t D_{t,s} A_t + d D_{t,s}))$ 
8:    $S_{t,s} \leftarrow -\rho \text{msgn}(R_{t,s})$ 
9:    $\Delta_{t,s} \leftarrow S_{t,s} - D_{t,s}$ 
10:   $g_{t,s}^{\text{FW}} \leftarrow \langle R_{t,s}, D_{t,s} - S_{t,s} \rangle$ 
11:   $c_{t,s} \leftarrow \tau(\alpha_t \text{tr}(\Delta_{t,s}^\top B_t \Delta_{t,s} A_t) + d \|\Delta_{t,s}\|_F^2)$ 
12:  if  $c_{t,s} > 0$  then
13:     $\gamma_{t,s} \leftarrow \text{clip}_{[0,1]} \left( \frac{g_{t,s}^{\text{FW}}}{\eta_t c_{t,s}} \right)$ 
14:  else if  $g_{t,s}^{\text{FW}} > 0$  then
15:     $\gamma_{t,s} \leftarrow 1$ 
16:  else
17:     $\gamma_{t,s} \leftarrow 0$ 
18:  end if
19:   $D_{t,s+1} \leftarrow D_{t,s} + \gamma_{t,s} \Delta_{t,s}$ 
20: end for
21:  $W_{t+1} \leftarrow W_t + \eta_t D_{t,K}$ 
```

4.1 Curvature Induces Nonuniform Singular Amplitudes

We first isolate the effect of curvature on singular amplitudes. Suppose the singular modes of the gradient are also eigenmodes of the curvature operator. Let

$$G = U \text{diag}(g_1, \dots, g_p) V^\top = \sum_{i=1}^p g_i u_i v_i^\top, \quad g_i \geq 0, \quad (19)$$

and define $E_i = u_i v_i^\top$. Assume $\mathcal{C}[E_i] = \kappa_i E_i$ with $\kappa_i \geq 0$. Under this aligned setting, curvature does not change the singular directions, but it can change the preferred magnitude of each mode.

Proposition 4.1 (Optimal spectrum under aligned curvature). *Consider*

$$\min_{\|D\|_2 \leq \rho} \left\{ \langle G, D \rangle + \frac{\eta}{2} \langle D, \mathcal{C}[D] \rangle \right\}, \quad (20)$$

under aligned curvature. Then there exists an optimal solution of the form

$$D^* = -U \text{diag}(s_1^*, \dots, s_p^*) V^\top, \quad (21)$$

where

$$s_i^* = \begin{cases} \rho, & \kappa_i = 0, \\ \min \left\{ \rho, \frac{g_i}{\eta \kappa_i} \right\}, & \kappa_i > 0. \end{cases} \quad (22)$$

The proof is given in Appendix A.1.

Muon sets every active singular value to ρ . In contrast, Proposition 4.1 shows that the curvature-aware optimum saturates mode i only when $g_i \geq \eta\rho\kappa_i$. Otherwise, $s_i^* = \frac{g_i}{\eta\kappa_i} < \rho$. Thus, even when gradient and curvature share the same singular directions, the quadratic optimum generally has a nonuniform singular-value profile. Intuitively, directions with larger curvature are more costly to move along, so they should receive smaller updates unless the gradient signal is sufficiently strong.

4.2 Curvature Can Change the Singular Directions

We next ask whether curvature can also change the singular directions of the optimal update. Muon preserves the singular directions of the gradient: if $G = U\Sigma V^\top$, then $D_M = -\rho UV^\top$. This property does not generally hold for the quadratic problem.

We first consider the case where the unconstrained quadratic minimizer is feasible.

Proposition 4.2 (Singular-direction adaptation under K-FAC). *Suppose*

$$\mathcal{C}[D] = BDA, \quad A \succ 0, \quad B \succ 0,$$

and let

$$G = g uv^\top, \quad g > 0, \quad \|u\|_2 = \|v\|_2 = 1.$$

If the unconstrained minimizer satisfies the spectral-norm constraint, then

$$D^* = -\frac{g}{\eta}(B^{-1}u)(A^{-1}v)^\top. \quad (23)$$

Hence, D^ preserves the left and right singular directions of G if and only if u and v are eigenvectors of B and A , respectively.*

The proof is given in Appendix A.2. Proposition 4.2 shows that anisotropic curvature can change both the left and right singular directions. This result, however, does not require the spectral-norm constraint to alter the unconstrained minimizer. We next show that direction adaptation also occurs when the constraint is active.

Proposition 4.3 (Direction adaptation with an active spectral constraint). *Suppose*

$$\mathcal{C}[D] = DA, \quad A \succ 0, \quad G = g uv^\top, \quad (24)$$

where $g > 0$ and $\|u\|_2 = \|v\|_2 = 1$. If the unconstrained minimizer lies outside the spectral-norm ball, then there exists a unique $\lambda > 0$ such that an optimal solution is

$$D^* = -g u [(\eta A + \lambda I)^{-1} v]^\top, \quad (25)$$

with

$$\|D^*\|_2 = \rho. \quad (26)$$

If v is not an eigenvector of A , then the right singular direction of D^ is not parallel to that of G .*

The proof is given in Appendix A.3. The multiplier $\lambda > 0$ appears because the spectral-norm constraint is binding. Nevertheless, $(\eta A + \lambda I)^{-1}v$ is generally not parallel to v . Therefore, the spectral-norm constraint does not force the quadratic optimum to preserve the singular directions of the gradient.

The intuition is that the spectral-norm constraint itself does not determine the singular structure of the update. Muon’s gradient-aligned directions arise from the linear objective, where only first-order descent is considered. With curvature, the optimal update balances descent and second-order cost, under anisotropic curvature, directions with larger curvature cost are typically downweighted unless they provide sufficiently stronger descent. The transformations B^{-1} and A^{-1} in the K-FAC solution explicitly capture this curvature-induced reweighting.

A simple two-dimensional example makes this explicit. Let

$$B = I, \quad A = \begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix}, \quad G = \begin{pmatrix} 101 & 2 \\ 0 & 0 \end{pmatrix}, \quad \eta = 1, \quad (27)$$

and choose

$$\rho = \sqrt{2}. \quad (28)$$

The unconstrained quadratic minimizer is

$$D_{\text{unc}} = -GA^{-1} = -\begin{pmatrix} 101/100 & 2 \\ 0 & 0 \end{pmatrix}. \quad (29)$$

Its spectral norm satisfies

$$\|D_{\text{unc}}\|_2 = \sqrt{\left(\frac{101}{100}\right)^2 + 4} > \sqrt{2} = \rho, \quad (30)$$

so the spectral-norm constraint is active.

The KKT conditions in Proposition 4.3 are satisfied with $\lambda = 1$, which gives

$$D^* = -G(A + I)^{-1} = -\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \|D^*\|_2 = \sqrt{2} = \rho. \quad (31)$$

The right singular direction of G is proportional to $(101, 2)^\top$, whereas that of D^* is proportional to $(1, 1)^\top$. Thus, the optimal singular direction changes even though the solution lies on the boundary of the same spectral-norm ball used by Muon.

4.3 QSD vs. Muon in Practice

The preceding analysis shows that curvature can change both the singular amplitudes and singular directions of the update. We now examine whether these effects occur in practice. At a fixed training step, we compute the Muon and QSD directions from the same momentum matrix $M_t = U\Sigma V^\top$, thereby excluding differences caused by training trajectories. For this analysis, Muon is computed using the exact polar factor.

We measure spectral and directional deviations by

$$\Delta_{\text{spec}}(D) = \frac{\|\sigma(D) - \bar{\sigma}(D)\mathbf{1}\|_2}{\|\sigma(D)\|_2}, \quad \Delta_{\text{dir}}(D; M) = \frac{\|D - \mathcal{P}_M(D)\|_F}{\|D\|_F}, \quad (32)$$

where $\mathcal{P}_M(D) = U \text{diag}(\text{diag}(U^\top D V)) V^\top$, and $\sigma(D) = (\sigma_1(D), \dots, \sigma_r(D))^\top$, $\bar{\sigma}(D) = r^{-1} \sum_{i=1}^r \sigma_i(D)$ is the mean singular value, and $\mathbf{1} \in \mathbb{R}^r$ is the all-ones vector.

Here, Δ_{spec} measures the relative departure from a flat singular spectrum: it is zero when all singular modes have equal amplitudes. Δ_{dir} measures the relative off-diagonal component of D in the singular basis of M : it is zero when D preserves the singular directions of M , and increases as different singular modes are mixed. Both quantities are zero for the exact Muon update.

Figure 1 shows that QSD differs from Muon in both aspects. Its singular spectrum is clearly nonuniform (Figure 1(a)), while $U^\top D_{\text{QSD}} V$ contains substantial off-diagonal mass (Figure 1(b)), indicating that QSD also changes the singular directions. Across layers, both deviations remain consistently nonzero (Figure 1(c)). These observations confirm that the curvature-induced effects identified above also arise during training.

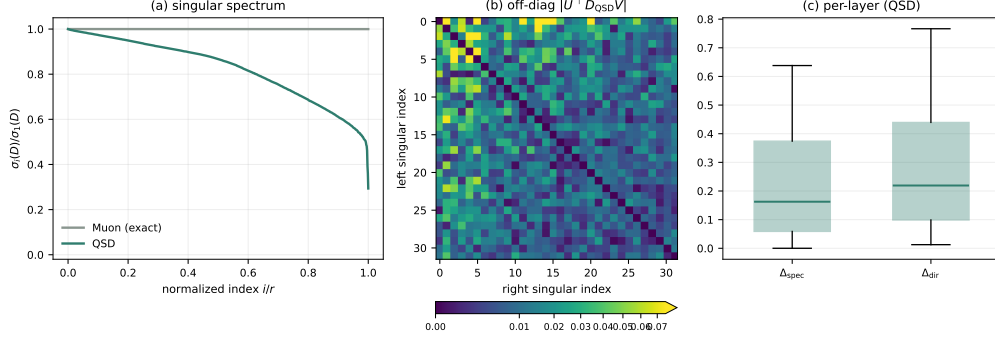


Figure 1: **QSD differs from Muon in both singular amplitudes and directions at a mild step of training.** (a) Normalized singular spectra of Muon and QSD for a representative layer. (b) Magnitude of $U^\top D_{\text{QSD}} V$ in the momentum singular direction (top 32×32 block). Off-diagonal entries indicate changes in singular directions. (c) Distribution of Δ_{spec} and Δ_{dir} across QSD layers.

5 Theoretical Guarantees

The QSD update is obtained by approximately solving a constrained quadratic problem, so it is important to understand both the quality of the inner solution and its relation to the original Muon update. We show that the Frank–Wolfe gap provides a computable certificate of surrogate suboptimality, exact line search decreases the surrogate monotonically, and the zero-initialized solver cannot be worse than Muon under the same quadratic surrogate. We then study the effect of curvature inflation and establish an $O(1/K)$ convergence rate for the practical QSD solver. All proofs are deferred to Appendix A.

5.1 Optimality Certificate and Monotonic Descent

Consider the convex quadratic surrogate

$$\hat{q}_\eta(D) = \langle G, D \rangle + \frac{\eta}{2} \langle D, \hat{\mathcal{C}}[D] \rangle, \quad \|D\|_2 \leq \rho, \quad (33)$$

where $\hat{\mathcal{C}}$ is positive semidefinite and self-adjoint. Let

$$\hat{D}^\star \in \arg \min_{\|D\|_2 \leq \rho} \hat{q}_\eta(D) \quad (34)$$

denote an optimal solution.

At Frank–Wolfe iteration s , define

$$R_s = \nabla_D \hat{q}_\eta(D_s) = G + \eta \hat{\mathcal{C}}[D_s]. \quad (35)$$

The linear minimization oracle is

$$S_s \in \arg \min_{\|S\|_2 \leq \rho} \langle R_s, S \rangle = -\rho \text{msgn}(R_s), \quad (36)$$

and the corresponding Frank–Wolfe gap is

$$\mathcal{G}_s = \langle R_s, D_s - S_s \rangle = \langle R_s, D_s \rangle + \rho \|R_s\|_*, \quad (37)$$

where the second equality follows from spectral–nuclear norm duality.

Proposition 5.1 (Frank–Wolfe optimality certificate). *For every feasible D_s ,*

$$0 \leq \hat{q}_\eta(D_s) - \hat{q}_\eta(\hat{D}^\star) \leq \mathcal{G}_s. \quad (38)$$

In particular, $\mathcal{G}_s = 0$ if and only if D_s is optimal.

The proof is given in Appendix A.4. Intuitively, $\mathcal{G}_s$ measures how much first-order improvement is still available within the spectral-norm ball. A small gap therefore indicates that no feasible direction can substantially improve the current surrogate value.

Exact line search also makes the surrogate decrease monotonically.

Proposition 5.2 (Monotonic surrogate descent). *Suppose*

$$\gamma_s \in \arg \min_{\gamma \in [0,1]} \hat{q}_\eta(D_s + \gamma(S_s - D_s)). \quad (39)$$

Then

$$\hat{q}_\eta(D_{s+1}) \leq \hat{q}_\eta(D_s), \quad (40)$$

with strict inequality whenever $\mathcal{G}_s > 0$.

The proof can be found in Appendix A.5. Because $\gamma = 0$ is feasible, exact line search cannot increase the surrogate. When $\mathcal{G}_s > 0$, the Frank–Wolfe direction is strictly descending.

5.2 Relation to the Muon Update

The zero-initialized solver can be compared directly with the Muon update.

Proposition 5.3 (One-step surrogate dominance over Muon). *Suppose that $D_0 = 0$ and exact line search is used. Let*

$$D_M = -\rho \operatorname{msgn}(G). \quad (41)$$

Then

$$\hat{q}_\eta(D_1) \leq \hat{q}_\eta(D_M). \quad (42)$$

Moreover, for every $K \geq 1$,

$$\hat{q}_\eta(D_K) \leq \hat{q}_\eta(D_M). \quad (43)$$

Since $D_0 = 0$, the first Frank–Wolfe atom is $S_0 = D_M$. Exact line search therefore chooses the best point on the segment

$$\{\gamma D_M : \gamma \in [0, 1]\}.$$

which contains the full Muon update at $\gamma = 1$. Subsequent Frank–Wolfe steps preserve the inequality by Proposition 5.2. The proof of Proposition 5.3 can be found in Appendix A.6.

This comparison concerns the quadratic surrogate rather than the exact training loss, and uses zero initialization only to make the relation to Muon explicit.

5.3 Effect of Curvature Inflation

We next examine the curvature inflation used in Section 3.3. Consider the aligned setting, and let $g_i \geq 0$ and $\kappa_i \geq 0$ denote the gradient magnitude and curvature of the i -th mode.

Proposition 5.4 (Effect of curvature inflation). *With calibration $\alpha > 0$, inflation $\tau \geq 1$, and damping $d \geq 0$, the optimal amplitude of the i -th mode is*

$$s_i^\star(\tau) = \min \left\{ \rho, \frac{g_i}{\eta\tau(\alpha\kappa_i + d)} \right\}. \quad (44)$$

For $g_i > 0$, define the effective curvature-to-gradient ratio

$$\chi_i = \frac{\eta\rho(\alpha\kappa_i + d)}{g_i}. \quad (45)$$

The mode remains saturated at the spectral-norm boundary if $\tau\chi_i \leq 1$, and becomes curvature-limited if $\tau\chi_i > 1$. Equivalently, the transition occurs at the critical curvature

$$\kappa_i^{\text{crit}}(\tau) = \frac{1}{\alpha} \left(\frac{g_i}{\eta\tau\rho} - d \right), \quad (46)$$

which decreases monotonically with τ :

$$\frac{\partial \kappa_i^{\text{crit}}}{\partial \tau} = -\frac{g_i}{\alpha\eta\rho\tau^2} < 0. \quad (47)$$

Once a mode is curvature-limited, its amplitude decreases as $1/\tau$.

The proof of Proposition 5.4 can be found in Appendix A.8. Intuitively, high-curvature directions are more sensitive to finite updates, since the same update magnitude induces a larger second-order change in the objective. Increasing τ therefore makes these sensitive directions more conservative while preserving less curvature-limited directions.

5.4 Convergence of the QSD Inner Solver

We now specialize to the practical QSD surrogate at a fixed training iteration t :

$$\hat{q}_t(D) = \langle M_t, D \rangle + \frac{\eta_t\tau}{2} [\alpha_t \text{tr}(D^\top B_t D A_t) + d\|D\|_F^2], \quad \|D\|_2 \leq \rho. \quad (48)$$

Assume

$$A_t \succeq 0, \quad B_t \succeq 0, \quad \alpha_t > 0, \quad d \geq 0, \quad (49)$$

and let

$$D_t^* \in \arg \min_{\|D\|_2 \leq \rho} \hat{q}_t(D) \quad (50)$$

denote an optimal solution.

Theorem 5.5 (Inner-loop convergence of QSD). *Let $D_{t,K}$ denote the iterate after $K \geq 1$ Frank–Wolfe refinements with exact line search, starting from any feasible initialization $D_{t,0}$. Then, with $r = \min\{m, n\}$,*

$$\hat{q}_t(D_{t,K}) - \hat{q}_t(D_t^*) \leq \frac{8\eta_t\tau r\rho^2 (\alpha_t\|A_t\|_2\|B_t\|_2 + d)}{K + 2}. \quad (51)$$

Hence, the inner solver converges to the optimal surrogate value at rate $O(K^{-1})$.

The proof is given in Appendix A.7. This matches the classical $O(1/K)$ convergence rate of Frank–Wolfe for smooth convex objectives [11], with the constant here specialized to the spectral-norm feasible set and the K-FAC quadratic curvature. Its constant increases with the size of the feasible spectral-norm ball and with the largest curvature of the practical quadratic, reflecting a harder inner optimization problem.

Because Theorem 5.5 holds for any feasible initialization, it also applies directly to the practical warm start $D_{t,0} = D_{t-1,K}$. When $d > 0$, the surrogate is additionally strongly convex, which implies convergence of the iterate itself to the unique inner optimum. The corresponding bound is given in Appendix B.

6 Experiments

In this section, we first compare QSD with Muon on the GPT architecture, and provide the results of the hyperparameter sensitivity experiment and training efficiency experiment of QSD. Then we compare the performance of QSD and the Muon variants.

6.1 Pre-training

Baseline. We first pre-train GPT-124M with a compute-optimal [10] token-to-parameter (T2P) ratio. All models are trained on the FineWeb dataset [22], with the same tokenizer as Muon Speed Run [12] and Newton-Muon [6], with a vocabulary size of 50257 and a batch size of 512. Training is conducted on RTX 5000 Ada GPUs using mixed precision (bfloat16). Following the setup in [12, 6, 34, 1], we apply QSD (or Muon) to all parameters except embeddings, unembeddings, normalization layers, and positional encodings, which are optimized using AdamW. For the polar operator, we adopt the same configuration as in [13]. We sweep the AdamW learning rate in $[0.001, 0.0018, 0.0032, 0.0056, 0.01]$ and the remaining part in $[0.005, 0.01, 0.02, 0.04]$ and report the best results in Table 1. Detailed hyperparameters are provided in the Appendix E. As shown in Table 1, QSD outperforms Muon at 6% time overhead.

Table 1: Validation performance and training time of pretraining GPT-124M on 4 RTX 5000 Ada GPUs.

Method	Run 1	Run 2	Run 3	Avg. loss	Avg. time (s)	Time Overhead (%)
Muon	3.3318	3.3312	3.3319	3.3316	7390.6	0.0
QSD	3.3151	3.3161	3.3161	3.3158 (-0.0158)	7833.9	+6.00

Ablation for baseline. We tune the QSD-specific hyperparameters on GPT-124M and reuse the resulting configuration for larger-scale experiments unless otherwise stated. Tables 2 and 3 show the sensitivity to the number of FW steps and K-FAC damping, respectively. Increasing the number of FW steps improves validation loss up to about three steps, after which the gains become marginal. For K-FAC damping, a moderate damping value of 10^{-6} performs best, while both larger and smaller damping slightly degrade validation loss.

Table 2: Effect of FW Steps.

FW step	1	2	3	4	5
Val. loss	3.3212	3.3199	3.3159	3.3161	3.3155

Table 3: Effect of K-FAC Damping.

d	1e-4	1e-6	1e-8
Val. loss	3.3199	3.3159	3.3182

We further study the effects of the update interval and EMA coefficient for K-FAC factor estimation and curvature-scale calibration. Figure 2a and Figure 2b show a broad region of stable performance, where most variations are small and likely comparable to training noise. However, overly frequent or infrequent updates combined with insufficient smoothing can lead to noticeable degradation.

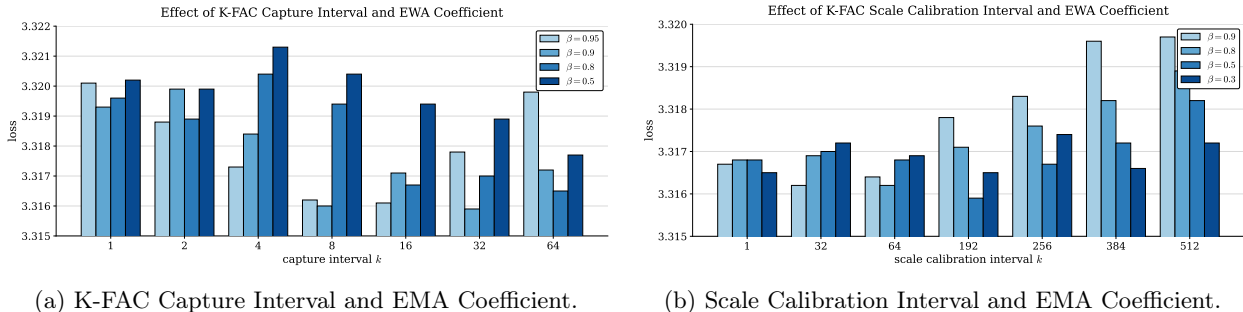


Figure 2: Sensitivity to K-FAC factor estimation and curvature-scale calibration settings.

Tables 4 and 5 examine the effects of the data sampling ratio used for K-FAC factor estimation and the sample ratio used for scale calibration. For K-FAC factor estimation, using only 1% of the data achieves the best validation loss, with little benefit from larger sampling ratios, and for scale calibration, performance

is stable across a wide range of sampling ratios, with 1% already matching or outperforming substantially larger ratios.

Table 4: Effect of K-FAC Sample Ratio.

Ratio	100%	10%	1%	0.5%	0.1%
Val. loss	3.3181	3.3161	3.3159	3.3173	3.3178

Table 5: K-FAC Scale Calibration Sample Ratio.

Ratio	40%	10%	1%	0.5%	0.1%
Val. loss	3.3177	3.3169	3.3159	3.3169	3.3171

Table 6 studies the effect of the curvature inflation factor. Performance improves as τ increases up to 1.5, suggesting that mild curvature inflation benefits training by making curvature-sensitive directions more conservative, while excessive inflation can over-regularize the update.

Table 6: Effect of K-FAC Inflation Factor.

τ	0.75	1.0	1.25	1.5	1.75	2.0
Val. loss	3.3193	3.3173	3.3166	3.3159	3.3168	3.3175

GPT-350M. To verify whether the performance advantage of QSD can be maintained at a larger scale, we pre-trained GPT-350M under the optimal T2P ratio. We reuse the QSD-specific hyperparameters tuned on GPT-124M, and only adjust the learning rate together with the K-FAC factor-refresh and calibration intervals (and their EMA coefficients) to account for the longer training schedule. All other settings are kept unchanged; full values are listed in Table 10. As shown in Table 7, QSD can still ensure performance gains after the model scale is increased.

Table 7: Validation performance and training time of pretraining GPT-350M on 4 RTX 5000 Ada GPUs.

Method	Val loss	Time (s)	Time Overhead (%)
Muon	3.0148	53569.3	0.0
QSD	3.0018 (-0.013)	56986.3	+6.38

6.2 Training Efficiency

In this section, we report the token and the wall-clock time required to reach the same validation loss, where the target is set to the final validation loss of the QSD. The results are shown in Table 8. QSD speeds up pretraining by up to 8.49% in the evaluated settings.

Table 8: Training cost required to reach a target loss.

Model	Target loss	Token Speed-up	Wall-clock Time Speed-up
GPT-124M	3.3159	$\uparrow$ 15%	$\uparrow$ 8.49%
GPT-350M	3.0018	$\uparrow$ 14.3%	$\uparrow$ 7.4%

6.3 Comparison with Muon Variants

In this section, we compare QSD with the other Muon variants, NorMuon [16] and the Newton-Muon [6]. As shown in Table 9, QSD consistently outperforms both NorMuon and Newton-Muon across all three runs.

7 Conclusion

We studied what happens to the Muon update when the linear local model is replaced by a quadratic one while keeping the same spectral-norm constraint. Our analysis shows that the flat singular spectrum and

Table 9: Validation performance of GPT-124M on different methods on 4 RTX 5000 Ada GPUs.

Method	Run 1	Run 2	Run 3	Avg. loss
NorMuon	3.3264	3.3280	3.3265	3.3270
Newton Muon	3.3233	3.3237	3.3241	3.3237
QSD	3.3151	3.3161	3.3161	3.3158

gradient-aligned singular directions of Muon are not imposed by the spectral-norm constraint itself: curvature can change both the singular values and the singular directions of the optimal update. We made this quadratic problem practical using Kronecker-factored curvature and a small number of Frank–Wolfe steps. The resulting QSD method admits simple optimization guarantees and consistently improves validation loss over Muon and recent Muon variants in GPT pre-training. At matched validation loss, QSD reduces wall-clock training time by up to 8.49% in our experiments.

8 Limitations

QSD relies on a local quadratic model, whose quality depends on the curvature approximation. In our practical implementation, the Kronecker factors are constructed from empirical-Fisher statistics and their overall scale is calibrated using GGN directional curvature. This calibration corrects scale mismatch along the calibration direction, but does not remove possible mismatch in the full curvature structure. The additional inflation factor τ is also selected empirically rather than derived from a bound on the finite-step approximation error.

Our optimization guarantees concern the constrained quadratic surrogate. Because the practical curvature is approximate and higher-order terms are not modeled, a better surrogate value does not by itself guarantee a larger decrease in the exact training loss. Finally, our experiments focus on GPT pre-training at the 124M and 350M scales. Evaluating QSD on substantially larger models, other architectures, and post-training settings would further establish when the additional curvature computation is worthwhile.

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A Proofs of Theoretical Results

This appendix provides proofs for the results stated in Sections 4 and 5.

A.1 Proof of Proposition 4.1

Let

$$\mathcal{S} = \text{span}\{E_1, \dots, E_p\}.$$

Because every E_i is an eigenmode of the self-adjoint operator $\mathcal{C}$, the subspace $\mathcal{S}$ is invariant under $\mathcal{C}$. Its orthogonal complement $\mathcal{S}^\perp$ is also invariant.

For any feasible D , write

$$D = D_\parallel + D_\perp, \quad D_\parallel \in \mathcal{S}, \quad D_\perp \in \mathcal{S}^\perp. \quad (52)$$

Since $G \in \mathcal{S}$,

$$\langle G, D_\perp \rangle = 0.$$

Moreover, self-adjointness and invariance imply

$$\langle D_\parallel, \mathcal{C}[D_\perp] \rangle = 0.$$

Therefore,

$$\begin{aligned} q_\eta(D) &= \langle G, D_\parallel \rangle + \frac{\eta}{2} \langle D_\parallel, \mathcal{C}[D_\parallel] \rangle \\ &\quad + \frac{\eta}{2} \langle D_\perp, \mathcal{C}[D_\perp] \rangle \\ &\geq q_\eta(D_\parallel), \end{aligned} \tag{53}$$

where the inequality follows from $\mathcal{C} \succeq 0$.

We next verify that $D_\parallel$ remains feasible. Write

$$D_\parallel = \sum_{i=1}^p d_i E_i, \quad d_i = \langle E_i, D \rangle = u_i^\top D v_i.$$

For every i ,

$$|d_i| = |u_i^\top D v_i| \leq \|D\|_2.$$

Since the matrices $E_i = u_i v_i^\top$ share orthonormal left and right singular vectors,

$$\|D_\parallel\|_2 = \max_i |d_i| \leq \|D\|_2 \leq \rho.$$

Thus $D_\parallel$ is feasible, and there exists an optimal solution entirely in $\mathcal{S}$.

We may therefore write

$$D = - \sum_{i=1}^p s_i E_i = -U \operatorname{diag}(s_1, \dots, s_p) V^\top. \tag{54}$$

Its spectral norm is

$$\|D\|_2 = \max_i |s_i|,$$

so feasibility is equivalent to

$$|s_i| \leq \rho \quad \text{for all } i.$$

Using

$$\mathcal{C}[E_i] = \kappa_i E_i$$

and the orthonormality of the E_i 's, the objective becomes

$$q_\eta(D) = \sum_{i=1}^p \left(-g_i s_i + \frac{\eta \kappa_i}{2} s_i^2 \right). \tag{55}$$

Because $g_i \geq 0$, an optimizer may be chosen with $s_i \geq 0$. The problem therefore decouples into

$$\min_{0 \leq s_i \leq \rho} \left\{ -g_i s_i + \frac{\eta \kappa_i}{2} s_i^2 \right\}. \tag{56}$$

If $\kappa_i > 0$, the unconstrained minimizer satisfies

$$-g_i + \eta \kappa_i s_i = 0,$$

and therefore

$$s_i^{\text{unc}} = \frac{g_i}{\eta \kappa_i}.$$

Clipping to the feasible interval gives

$$s_i^* = \min \left\{ \rho, \frac{g_i}{\eta \kappa_i} \right\}.$$

If $\kappa_i = 0$, the mode-wise objective becomes $-g_i s_i$, which is minimized at $s_i^* = \rho$ whenever $g_i > 0$. If $g_i = 0$, every $s_i \in [0, \rho]$ is optimal, so $s_i^* = \rho$ may be selected without loss of generality. This proves Proposition 4.1.

A.2 Proof of Proposition 4.2

Proof. For

$$q_\eta(D) = \langle G, D \rangle + \frac{\eta}{2} \langle D, BDA \rangle,$$

the first-order optimality condition for the unconstrained problem is

$$G + \eta BDA = 0. \quad (57)$$

Since $A \succ 0$ and $B \succ 0$, both factors are invertible, and the unique unconstrained minimizer is therefore

$$D_{\text{unc}} = -\frac{1}{\eta} B^{-1} G A^{-1}. \quad (58)$$

Substituting $G = g u v^\top$ gives

$$D_{\text{unc}} = -\frac{g}{\eta} B^{-1} u v^\top A^{-1} \quad (59)$$

$$= -\frac{g}{\eta} (B^{-1} u) (A^{-1} v)^\top, \quad (60)$$

where we used the symmetry of A . By assumption, this unconstrained minimizer lies inside the spectral-norm ball, so it is also the solution of the constrained problem.

Because $D^\star$ is rank one, its left and right singular directions are proportional to $B^{-1}u$ and $A^{-1}v$, respectively. Hence, $D^\star$ has the same singular directions as $G = g u v^\top$ if and only if

$$B^{-1}u \parallel u, \quad A^{-1}v \parallel v. \quad (61)$$

For an invertible symmetric matrix, $B^{-1}u \parallel u$ holds if and only if u is an eigenvector of B , and similarly $A^{-1}v \parallel v$ holds if and only if v is an eigenvector of A . This proves the claim. $\square$

A.3 Proof of Proposition 4.3

Proof. Consider

$$\min_{\|D\|_2 \leq \rho} \left\{ \langle G, D \rangle + \frac{\eta}{2} \text{tr}(D^\top D A) \right\}, \quad (62)$$

where

$$G = g u v^\top, \quad A \succ 0, \quad \|u\|_2 = \|v\|_2 = 1.$$

For any feasible D , decompose

$$D = u x^\top + D_\perp, \quad u^\top D_\perp = 0. \quad (63)$$

Since $G = g u v^\top$,

$$\langle G, D_\perp \rangle = 0. \quad (64)$$

Moreover,

$$\begin{aligned} \text{tr}(D^\top D A) &= x^\top A x + \text{tr}(D_\perp^\top D_\perp A) \\ &\geq x^\top A x, \end{aligned} \quad (65)$$

because $A \succ 0$. Also,

$$\|u x^\top\|_2 = \|x\|_2 \leq \|D\|_2. \quad (66)$$

Therefore, there exists an optimal solution of the form

$$D^\star = u x^{\star\top}, \quad (67)$$

where $x^\star$ solves

$$\min_{\|x\|_2 \leq \rho} \left\{ g v^\top x + \frac{\eta}{2} x^\top A x \right\}. \quad (68)$$

The unconstrained minimizer is

$$x_{\text{unc}} = -\frac{g}{\eta} A^{-1} v. \quad (69)$$

By assumption,

$$\|x_{\text{unc}}\|_2 > \rho, \quad (70)$$

so the Euclidean-norm constraint in Eq. (68) is active.

The KKT conditions are therefore

$$g v + \eta A x^\star + \lambda x^\star = 0, \quad \lambda > 0, \quad \|x^\star\|_2 = \rho. \quad (71)$$

Solving the stationarity condition gives

$$x^\star = -g(\eta A + \lambda I)^{-1} v. \quad (72)$$

Hence,

$$D^\star = -g u [(\eta A + \lambda I)^{-1} v]^\top, \quad (73)$$

with $\lambda > 0$ chosen so that $\|D^\star\|_2 = \|x^\star\|_2 = \rho$.

It remains to compare the right singular directions. Since G is rank one, its right singular direction is v . The right singular direction of $D^\star$ is proportional to $(\eta A + \lambda I)^{-1} v$.

If these two directions were parallel, then for some scalar $c \neq 0$,

$$(\eta A + \lambda I)^{-1} v = c v. \quad (74)$$

Multiplying by $\eta A + \lambda I$ gives

$$v = c(\eta A + \lambda I) v, \quad (75)$$

and therefore

$$A v = \frac{1/c - \lambda}{\eta} v. \quad (76)$$

Thus, v would be an eigenvector of A .

Consequently, if v is not an eigenvector of A ,

$$(\eta A + \lambda I)^{-1} v \not\parallel v, \quad (77)$$

and the right singular direction of the constrained optimum differs from that of the gradient. Since $\lambda > 0$ and $\|D^\star\|_2 = \rho$, this direction change occurs while the spectral-norm constraint is active. $\square$

A.4 Proof of Proposition 5.1

Proof. Since $\hat{q}_\eta$ is convex, for any feasible D ,

$$\hat{q}_\eta(D) \geq \hat{q}_\eta(D_s) + \langle \nabla_D \hat{q}_\eta(D_s), D - D_s \rangle. \quad (78)$$

Taking $D = \hat{D}^\star$ and using $R_s = \nabla_D \hat{q}_\eta(D_s)$ gives

$$\hat{q}_\eta(D_s) - \hat{q}_\eta(\hat{D}^\star) \leq \langle R_s, D_s - \hat{D}^\star \rangle. \quad (79)$$

By definition, S_s minimizes the linearized objective over the same feasible set, and therefore

$$\langle R_s, S_s \rangle \leq \langle R_s, \hat{D}^* \rangle. \quad (80)$$

Hence,

$$\hat{q}_\eta(D_s) - \hat{q}_\eta(\hat{D}^*) \leq \langle R_s, D_s - \hat{D}^* \rangle \quad (81)$$

$$\leq \langle R_s, D_s - S_s \rangle \quad (82)$$

$$= \mathcal{G}_s. \quad (83)$$

The lower bound follows from the optimality of $\hat{D}^*$.

It remains to characterize the zero-gap case. Since $\mathcal{G}_s \geq 0$, if $\mathcal{G}_s = 0$, the bound above implies

$$\hat{q}_\eta(D_s) = \hat{q}_\eta(\hat{D}^*), \quad (84)$$

so D_s is optimal. Conversely, if D_s is optimal, the first-order optimality condition for convex optimization over the feasible set is

$$\langle R_s, D - D_s \rangle \geq 0 \quad \text{for all feasible } D. \quad (85)$$

In particular, taking $D = S_s$ gives

$$\langle R_s, D_s - S_s \rangle \leq 0. \quad (86)$$

Since the Frank–Wolfe gap is always nonnegative, this implies $\mathcal{G}_s = 0$. $\square$

A.5 Proof of Proposition 5.2

Proof. Let

$$\Delta_s = S_s - D_s, \quad (87)$$

and define the one-dimensional function

$$\phi_s(\gamma) = \hat{q}_\eta(D_s + \gamma\Delta_s), \quad \gamma \in [0, 1]. \quad (88)$$

Since $\gamma = 0$ is feasible for the line search and γ_s minimizes ϕ_s over $[0, 1]$,

$$\hat{q}_\eta(D_{s+1}) = \phi_s(\gamma_s) \leq \phi_s(0) = \hat{q}_\eta(D_s). \quad (89)$$

For strict descent, the directional derivative at $\gamma = 0$ is

$$\phi'_s(0) = \langle R_s, S_s - D_s \rangle \quad (90)$$

$$= -\mathcal{G}_s. \quad (91)$$

If $\mathcal{G}_s > 0$, then $\phi'_s(0) < 0$. By differentiability, there exists a sufficiently small $\gamma > 0$ such that

$$\phi_s(\gamma) < \phi_s(0). \quad (92)$$

Exact line search therefore satisfies

$$\hat{q}_\eta(D_{s+1}) < \hat{q}_\eta(D_s). \quad (93)$$

$\square$

A.6 Proof of Proposition 5.3

Proof. Under the zero initialization $D_0 = 0$,

$$R_0 = \nabla_D \hat{q}_\eta(0) = G. \quad (94)$$

The first Frank–Wolfe atom is therefore

$$S_0 = -\rho \operatorname{msgn}(G) = D_M. \quad (95)$$

The first iterate has the form

$$D_1 = D_0 + \gamma_0(S_0 - D_0) = \gamma_0 D_M, \quad (96)$$

where exact line search chooses

$$\gamma_0 \in \arg \min_{\gamma \in [0,1]} \hat{q}_\eta(\gamma D_M). \quad (97)$$

Since $\gamma = 1$ is feasible for this one-dimensional optimization,

$$\hat{q}_\eta(D_1) = \min_{\gamma \in [0,1]} \hat{q}_\eta(\gamma D_M) \leq \hat{q}_\eta(D_M). \quad (98)$$

This proves the one-step comparison.

For every subsequent Frank–Wolfe refinement, Proposition 5.2 gives

$$\hat{q}_\eta(D_K) \leq \hat{q}_\eta(D_{K-1}) \leq \cdots \leq \hat{q}_\eta(D_1). \quad (99)$$

Combining the two inequalities yields

$$\hat{q}_\eta(D_K) \leq \hat{q}_\eta(D_M), \quad K \geq 1. \quad (100)$$

□

A.7 Proof of Theorem 5.5

Proof. Fix a training iteration t . The practical QSD surrogate is

$$\hat{q}_t(D) = \langle M_t, D \rangle + \frac{\eta_t \tau}{2} [\alpha_t \operatorname{tr}(D^\top B_t D A_t) + d \|D\|_F^2]. \quad (101)$$

Its gradient is

$$\nabla_D \hat{q}_t(D) = M_t + \eta_t \tau (\alpha_t B_t D A_t + d D). \quad (102)$$

Define

$$L_t = \eta_t \tau (\alpha_t \|A_t\|_2 \|B_t\|_2 + d). \quad (103)$$

For any matrices D and D' ,

$$\begin{aligned} & \|\nabla_D \hat{q}_t(D) - \nabla_D \hat{q}_t(D')\|_F \\ &= \eta_t \tau \|\alpha_t B_t (D - D') A_t + d(D - D')\|_F \\ &\leq \eta_t \tau (\alpha_t \|B_t\|_2 \|A_t\|_2 + d) \|D - D'\|_F \\ &= L_t \|D - D'\|_F. \end{aligned} \quad (104)$$

Thus, $\hat{q}_t$ has an L_t -Lipschitz gradient with respect to the Frobenius norm.

Let

$$\mathcal{D} = \{D : \|D\|_2 \leq \rho\}. \quad (105)$$

For any $D \in \mathcal{D}$, with $r = \min\{m, n\}$,

$$\|D\|_F \leq \sqrt{r} \|D\|_2 \leq \sqrt{r} \rho. \quad (106)$$

Hence, for any $D, D' \in \mathcal{D}$,

$$\|D - D'\|_F \leq 2\sqrt{r} \rho. \quad (107)$$

The Frank–Wolfe curvature constant of $\hat{q}_t$ over $\mathcal{D}$ therefore satisfies

$$C_t \leq L_t \text{diam}_F(\mathcal{D})^2. \quad (108)$$

Using Eq. (107),

$$C_t \leq 4L_t r \rho^2 \quad (109)$$

$$= 4\eta_t \tau r \rho^2 (\alpha_t \|A_t\|_2 \|B_t\|_2 + d). \quad (110)$$

Let

$$h_s = \hat{q}_t(D_{t,s}) - \hat{q}_t(D_t^*). \quad (111)$$

The standard Frank–Wolfe descent inequality gives, for any $\gamma \in [0, 1]$,

$$h_{s+1} \leq (1 - \gamma)h_s + \frac{\gamma^2}{2} C_t. \quad (112)$$

Exact line search performs at least as well as any prescribed γ .

At the first iteration, choosing $\gamma = 1$ gives

$$h_1 \leq \frac{C_t}{2}. \quad (113)$$

For subsequent iterations, the standard Frank–Wolfe [11] argument with $\gamma_s = 2/(s + 2)$ yields

$$h_K \leq \frac{2C_t}{K + 2}, \quad K \geq 1. \quad (114)$$

Substituting Eq. (110) gives

$$\boxed{\hat{q}_t(D_{t,K}) - \hat{q}_t(D_t^*) \leq \frac{8\eta_t \tau r \rho^2 (\alpha_t \|A_t\|_2 \|B_t\|_2 + d)}{K + 2}.} \quad (115)$$

This proves the result. $\square$

A.8 Proof of Proposition 5.4

In the aligned setting, write

$$G = U \text{diag}(g_1, \dots, g_r) V^\top, \quad g_i \geq 0, \quad (116)$$

and suppose that the curvature operator shares the same matrix modes,

$$\mathcal{C}[u_i v_i^\top] = \kappa_i u_i v_i^\top, \quad \kappa_i \geq 0. \quad (117)$$

After calibration, damping, and curvature inflation, the effective curvature operator is

$$\tilde{\mathcal{C}}[D] = \tau (\alpha \mathcal{C}[D] + dD). \quad (118)$$

An aligned descent direction can be written as

$$D = -U \text{diag}(s_1, \dots, s_r) V^\top, \quad 0 \leq s_i \leq \rho. \quad (119)$$

Because the singular vectors are orthonormal, $\|D\|_2 \leq \rho$ is equivalent to $0 \leq s_i \leq \rho$ for every mode. Substituting Eq. (119) into the quadratic surrogate gives

$$q_\eta(D) = \langle G, D \rangle + \frac{\eta}{2} \langle D, \tilde{\mathcal{C}}[D] \rangle \quad (120)$$

$$= \sum_{i=1}^r \left[-g_i s_i + \frac{\eta\tau}{2} (\alpha\kappa_i + d) s_i^2 \right]. \quad (121)$$

Hence the optimization separates across modes. For each i , we solve

$$\min_{0 \leq s_i \leq \rho} \left\{ -g_i s_i + \frac{\eta\tau}{2} (\alpha\kappa_i + d) s_i^2 \right\}. \quad (122)$$

When $\alpha\kappa_i + d > 0$, the unconstrained minimizer satisfies

$$-g_i + \eta\tau(\alpha\kappa_i + d)s_i = 0, \quad (123)$$

and is therefore

$$\bar{s}_i = \frac{g_i}{\eta\tau(\alpha\kappa_i + d)}. \quad (124)$$

Restricting this minimizer to the feasible interval $[0, \rho]$ yields

$$s_i^*(\tau) = \min \left\{ \rho, \frac{g_i}{\eta\tau(\alpha\kappa_i + d)} \right\}, \quad (125)$$

which proves Eq. (44). If $\alpha\kappa_i + d = 0$ and $g_i > 0$, the scalar objective is linear and is minimized at $s_i^* = \rho$, consistent with the limiting form above.

For $g_i > 0$, the mode remains saturated at the boundary precisely when

$$\frac{g_i}{\eta\tau(\alpha\kappa_i + d)} \geq \rho. \quad (126)$$

Rearranging gives

$$\tau \frac{\eta\rho(\alpha\kappa_i + d)}{g_i} \leq 1, \quad (127)$$

or equivalently

$$\tau\chi_i \leq 1, \quad (128)$$

where χ_i is defined in Eq. (45). Hence $s_i^* = \rho$ throughout this regime, so changes in τ do not affect the amplitude as long as the boundary condition remains satisfied.

The same condition can be expressed as a threshold on curvature:

$$\kappa_i \leq \frac{1}{\alpha} \left(\frac{g_i}{\eta\tau\rho} - d \right) = \kappa_i^{\text{crit}}(\tau). \quad (129)$$

Differentiating with respect to τ gives

$$\frac{\partial \kappa_i^{\text{crit}}}{\partial \tau} = -\frac{g_i}{\alpha\eta\rho\tau^2} < 0. \quad (130)$$

Thus increasing τ monotonically lowers the maximum curvature for which a mode can remain boundary-saturated. If $\kappa_i^{\text{crit}}(\tau) < 0$, no mode with $\kappa_i \geq 0$ and the corresponding g_i can satisfy the boundary condition.

Once the mode lies in the curvature-limited regime,

$$s_i^*(\tau) = \frac{g_i}{\eta\tau(\alpha\kappa_i + d)} = \frac{1}{\tau} \frac{g_i}{\eta(\alpha\kappa_i + d)}. \quad (131)$$

Therefore,

$$\frac{\partial s_i^*}{\partial \tau} = -\frac{g_i}{\eta \tau^2 (\alpha \kappa_i + d)} < 0 \quad (132)$$

whenever $g_i > 0$, so inflation monotonically attenuates every curvature-limited mode.

Consequently, curvature inflation has two effects: it leaves boundary-saturated modes unchanged, while lowering the transition threshold at which modes become controlled by the quadratic penalty. For fixed g_i , modes with larger κ_i cross this threshold earlier as τ increases. This completes the proof.

B Iterate Convergence under Damping

For any perturbation H , the quadratic form induced by the Hessian of $\hat{q}_t$ is

$$\langle H, \nabla^2 \hat{q}_t[H] \rangle = \eta_t \tau [\alpha_t \text{tr}(H^\top B_t H A_t) + d \|H\|_F^2]. \quad (133)$$

Since $A_t \succeq 0$, $B_t \succeq 0$, and $\alpha_t > 0$,

$$\alpha_t \text{tr}(H^\top B_t H A_t) \geq 0. \quad (134)$$

Therefore, when $d > 0$,

$$\langle H, \nabla^2 \hat{q}_t[H] \rangle \geq \eta_t \tau d \|H\|_F^2. \quad (135)$$

Hence, $\hat{q}_t$ is $\eta_t \tau d$ -strongly convex with respect to the Frobenius norm, and its constrained minimizer D_t^* is unique.

Because D_t^* minimizes $\hat{q}_t$ over the convex feasible set, strong convexity implies

$$\hat{q}_t(D) - \hat{q}_t(D_t^*) \geq \frac{\eta_t \tau d}{2} \|D - D_t^*\|_F^2. \quad (136)$$

In particular,

$$\|D_{t,K} - D_t^*\|_F^2 \leq \frac{2 [\hat{q}_t(D_{t,K}) - \hat{q}_t(D_t^*)]}{\eta_t \tau d}. \quad (137)$$

Applying Theorem 5.5 gives

$$\|D_{t,K} - D_t^*\|_F^2 \leq \frac{2}{\eta_t \tau d} \frac{8\eta_t \tau r \rho^2 (\alpha_t \|A_t\|_2 \|B_t\|_2 + d)}{K + 2} \quad (138)$$

$$= \frac{16r \rho^2 (\alpha_t \|A_t\|_2 \|B_t\|_2 + d)}{d(K + 2)}. \quad (139)$$

Taking square roots yields

$$\boxed{\|D_{t,K} - D_t^*\|_F \leq 4\rho \sqrt{\frac{r (\alpha_t \|A_t\|_2 \|B_t\|_2 + d)}{d(K + 2)}}.} \quad (140)$$

C K-FAC Factor Estimation Details

We estimate the K-FAC factors in Eqs. (13)–(14) from a small subset of token positions in the current training minibatch. The same sampled positions are used for the activation and output-gradient statistics so that each pair $(a_{i,\ell}, \delta_{i,\ell})$ corresponds to the same token.

C.1 Token Sampling

Factor statistics are sampled at the token level. For each microbatch containing N token positions, we uniformly sample without replacement

$$n_{\text{keep}} = \max \{1, \lceil r_F N \rceil\} \quad (141)$$

positions, where r_F is the factor-sampling ratio.

In our default setting $r_F = 0.01$, so each factor estimate uses approximately 1% of the 524288 tokens in an optimizer step (5248 sampled positions). The per-microbatch counts depend on the per-device batch size and the number of devices; under data-parallel training the per-device sufficient statistics $\sum aa^\top$, $\sum \delta\delta^\top$ and the sample counts are summed across devices before normalization, so the pooled estimate is unaffected by the device count.

The factors are averaged over the number of sampled token positions. The same sampling indices are used for $A_{t,\ell}$ and $B_{t,\ell}$ to preserve the correspondence between activations and their output gradients.

C.2 Output-Gradient Rescaling

The output gradients used to construct $B_{t,\ell}$ are obtained from backward hooks. Their raw values contain the scaling introduced by mean-loss reduction, gradient accumulation, AMP loss scaling, and any additional loss scaling used during training. We restore the corresponding per-token scale before computing the K-FAC statistics.

Let $\delta_{i,\ell}^{\text{hook}}$ denote the output gradient captured by the backward hook. The gradient used in the factor estimate is

$$\delta_{i,\ell} = \frac{N n_{\text{accum}}}{s_{\text{AMP}} s_{\text{custom}}} \delta_{i,\ell}^{\text{hook}}, \quad (142)$$

where N is the number of token positions in the original microbatch, n_{accum} is the number of accumulated microbatches, s_{AMP} is the AMP loss-scaling factor, and s_{custom} denotes any additional multiplicative loss scaling. We set $s_{\text{custom}} = 1$ when no additional scaling is used.

Importantly, N in Eq. (142) is the token count before K-FAC subsampling. Subsampling changes only the number of terms used to estimate the factors and does not change the scale of the underlying per-token gradients.

The output-side factor is then computed as

$$\hat{B}_{t,\ell} = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \delta_{i,\ell} \delta_{i,\ell}^\top, \quad (143)$$

using the same sampled token positions as the corresponding activation-side factor.

C.3 Factor Refresh and EMA

The K-FAC factors are refreshed every $j = 32$ optimizer steps. At a refresh step, the current estimates are incorporated using

$$A_{t,\ell} = \beta_F A_{t-j,\ell} + (1 - \beta_F) \hat{A}_{t,\ell}, \quad B_{t,\ell} = \beta_F B_{t-j,\ell} + (1 - \beta_F) \hat{B}_{t,\ell}, \quad (144)$$

with $\beta_F = 0.9$.

At the first refresh, the running factors are initialized directly from the first estimates,

$$A_{t,\ell} = \hat{A}_{t,\ell}, \quad B_{t,\ell} = \hat{B}_{t,\ell}. \quad (145)$$

Since the EMA is not initialized from zero, no bias correction is used. Between refresh steps, the most recently stored factors are reused.

C.4 Numerical Precision

The sampled activations and output gradients follow the training precision and are stored in bfloat16. The matrix products used to form the K-FAC statistics are therefore computed from bfloat16 operands using the corresponding tensor-core GEMM path. Accumulation across microbatches is performed in FP32, and the resulting K-FAC factors and their EMA states are stored in FP32.

D Directional Curvature Calibration

K-FAC can have a scale mismatch with the corresponding generalized Gauss–Newton (GGN) curvature. We correct this mismatch with a per-layer scalar coefficient by matching the two directional curvatures along an accepted update direction.

Let $\bar{D}_{t,\ell}$ denote the stored update direction used for calibration for layer ℓ , and let θ_t be the parameter point associated with this direction. Both directional curvatures below are evaluated at the same parameter point θ_t .

D.1 Directional Curvature Matching

The K-FAC directional curvature along $\bar{D}_{t,\ell}$ is

$$\begin{aligned} c_{t,\ell}^{\text{KFAC}} &= \langle \bar{D}_{t,\ell}, B_{t,\ell} \bar{D}_{t,\ell} A_{t,\ell} \rangle \\ &= \text{tr} (\bar{D}_{t,\ell}^\top B_{t,\ell} \bar{D}_{t,\ell} A_{t,\ell}). \end{aligned} \quad (146)$$

Damping is excluded from this calculation. Thus, $\alpha_{t,\ell}$ calibrates only the Kronecker-factored term $B_{t,\ell} D A_{t,\ell}$, while the damping term dD and the inflation factor τ are applied separately.

For the GGN curvature, let $z_i(\theta)$ denote the logits for token i , and let $J_{i,t,\ell}$ be the Jacobian of z_i with respect to the parameters of layer ℓ at θ_t . For softmax cross-entropy,

$$H_{i,t} = \text{diag}(p_{i,t}) - p_{i,t} p_{i,t}^\top, \quad p_{i,t} = \text{softmax}(z_i(\theta_t)), \quad (147)$$

is the Hessian with respect to the logits. Define

$$v_{i,t,\ell} = J_{i,t,\ell} \text{vec}(\bar{D}_{t,\ell}). \quad (148)$$

The corresponding GGN directional curvature is

$$c_{t,\ell}^{\text{GGN}} = \frac{1}{N_t^{\text{cal}}} \sum_{i \in \mathcal{T}_t^{\text{cal}}} v_{i,t,\ell}^\top H_{i,t} v_{i,t,\ell}, \quad (149)$$

where $\mathcal{T}_t^{\text{cal}}$ contains all tokens from the sequences selected for calibration and

$$N_t^{\text{cal}} = |\mathcal{T}_t^{\text{cal}}|. \quad (150)$$

The GGN estimate and the K-FAC statistics use the same per-token averaging convention.

The instantaneous calibration coefficient is

$$\hat{\alpha}_{t,\ell} = \frac{c_{t,\ell}^{\text{GGN}}}{c_{t,\ell}^{\text{KFAC}}}. \quad (151)$$

Hence, before clipping and temporal smoothing, $\hat{\alpha}_{t,\ell}$ matches the GGN curvature along $\bar{D}_{t,\ell}$ without changing the Kronecker structure of $A_{t,\ell}$ and $B_{t,\ell}$.

Layers with

$$c_{t,\ell}^{\text{KFAC}} \leq \epsilon_{\text{skip}} \quad (152)$$

are skipped for that calibration step and retain their previous calibration coefficient, where

$$\epsilon_{\text{skip}} = 10^{-20}. \quad (153)$$

The instantaneous ratio is clipped before the EMA update:

$$\alpha_{t,\ell} = \beta_\alpha \alpha_{t-k,\ell} + (1 - \beta_\alpha) \text{clip}(\hat{\alpha}_{t,\ell}, \alpha_{\min}, \alpha_{\max}). \quad (154)$$

Between calibration steps, the most recent $\alpha_{t,\ell}$ is reused.

D.2 Finite-Difference GGN Estimate

We do not form the GGN matrix or the Jacobian explicitly. Only the Jacobian–vector product in Eq. (148) is needed, and we approximate it using a forward finite difference.

We first normalize the calibration direction,

$$\tilde{D}_{t,\ell} = \frac{\bar{D}_{t,\ell}}{\|\bar{D}_{t,\ell}\|_F}. \quad (155)$$

Let $\mathcal{E}_\ell[D]$ denote a perturbation that changes only layer ℓ by D . Then

$$\begin{aligned} v_{i,t,\ell} &= J_{i,t,\ell} \text{vec}(\tilde{D}_{t,\ell}) \\ &\approx \frac{\|\bar{D}_{t,\ell}\|_F}{\delta} \left[z_i(\theta_t + \delta \mathcal{E}_\ell[\tilde{D}_{t,\ell}]) - z_i(\theta_t) \right]. \end{aligned} \quad (156)$$

Substituting Eq. (156) into Eq. (149) gives the GGN directional curvature using only forward evaluations and the closed-form logit-space Hessian in Eq. (147). No explicit Jacobian, GGN matrix, or additional backward pass is required.

The direction and curvature are evaluated at the same parameter point. In the implementation, the parameters are restored to the θ_t associated with $\bar{D}_{t,\ell}$ before computing the unperturbed and perturbed logits. The unperturbed logits $z_i(\theta_t)$ are computed once and shared across the layerwise calibration calculations.

D.3 Implementation Details

We perform curvature calibration every

$$k = 192 \quad (157)$$

training iterations and use

$$\beta_\alpha = 0.5, \quad \alpha_{\min} = 0.05, \quad \alpha_{\max} = 100. \quad (158)$$

Calibration data are sampled at the sequence level rather than at the token level. We nominally sample 1% of the sequences in the current minibatch. A minimum calibration block size of 4 sequences is used, and the selected sequences are processed in blocks of 4. With the minibatch size of 512 used in our experiments, the nominal 1% setting results in 8 sequences per calibration step, corresponding to 1.56% of the minibatch. All tokens in each selected sequence are included in the directional-curvature estimate.

For the finite-difference approximation, we use

$$\delta = 0.1 \quad (159)$$

after normalizing the perturbation by $\|\bar{D}_{t,\ell}\|_F$. The finite-difference computation is carried out in FP32 with TF32 disabled. This is important because Eq. (156) subtracts two nearby logit vectors and is sensitive to reduced mantissa precision.

E Experimental Details

Table 10 summarizes the training and QSD-specific hyperparameters used in our main experiments. Unless otherwise stated, QSD-specific hyperparameters tuned on GPT-124M are kept unchanged for GPT-350M.

Table 10: Hyperparameters used in the main experiments.

Hyperparameter	GPT-124M	GPT-350M
<i>Training setup</i>		
Dataset	FineWeb	FineWeb
Sequence length	1024	1024
Batch size (sequences)	512	512
Training tokens	2.5B	7B
Precision	bfloat16	bfloat16
Number of GPUs	4	4
GPU	RTX 5000 Ada	RTX 5000 Ada
Learning-rate schedule	WSD	WSD
Warmup steps / ratio	0	0
Weight decay	0	0
<i>QSD optimizer</i>		
QSD momentum	0.95	0.95
QSD learning rate	0.02	0.015
Spectral radius ρ	1	1
Number of FW steps K	3	3
K-FAC damping d	10^{-6}	10^{-6}
Curvature inflation τ	1.5	1.5
<i>Muon optimizer</i>		
Muon momentum	0.95	0.95
Muon learning rate	0.01	0.01
Spectral radius ρ	1	1
<i>K-FAC factor estimation</i>		
Factor sampling ratio r_F	1%	1%
Factor refresh interval j	32 optimizer steps	128 optimizer steps
Factor EMA β_F	0.9	0.8
Factor storage / EMA precision	FP32	FP32
<i>Curvature-scale calibration</i>		
Calibration interval k	192 optimizer steps	384 optimizer steps
Calibration EMA β_α	0.5	0.5
Calibration clip	[0.05, 100]	[0.05, 100]
Nominal sequence sampling ratio	1%	1%
Finite-difference step δ	0.1	0.1
Finite-difference precision	FP32 (TF32 disabled)	FP32 (TF32 disabled)
<i>AdamW parameters</i>		
AdamW learning rate	0.0056	0.0032
AdamW β_1	0.9	0.9
AdamW β_2	0.95	0.95